Using Chaos Theory To Understand the Motion of a Particle in Convection Currents

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Abstract

Turbulence, the unpredictable motion of a fluid characterized by vortices and frequent velocity changes, has been called “the most important unsolved problem in physics.” Being able to predict turbulent flow would have massive implications in aerodynamics, energy, and medicine and would be one of the most important scientific discoveries ever. The goal of this project was to try to understand turbulent motion by seeing if the position-time data of a particle moving in convection currents would form a strange attractor in phase space, a strange attractor being a mathematical shape that reveals order in a system. The strange attractor would then be analyzed using chaotic methods to find underlying patterns in the seemingly random motion. The experiment involved releasing tapioca seeds into boiling water in a glass pot to create a controlled turbulent environment through convection currents. Particle motion was recorded in three dimensions from two angles using smartphones to capture three-dimensional movement. Then, the footage was analyzed using the OpenCV Python library to collect position and velocity data. The data was plotted in three-dimensional phase space using X position, Y position, and total velocity as axes. Despite conducting thirty trials, the resulting phase space plots showed no discernible pattern, suggesting that either the system will not show the hypothesized strange attractor pattern or experimental error prevented it from forming. Technical problems that limited the project include the size of the pot, the fact that tapioca seeds sink, and data loss due to motion-tracking software. Future experimentation will be conducted with improved experimental design with the goal of obtaining conclusive data.

1. Introduction

This experiment centers around understanding turbulence. When fluids flow in perfectly uniform sheets with uniform velocity, such as in **Figure 1**, it is called laminar flow. This type of fluid flow is simple to understand and to predict. However, when the sheets of uniform velocity mix, causing vortexes, eddies, and frequent changes in velocity in the fluid, this is called turbulent flow (OpenStax, n.d.). Perfect laminar flow is very hard to come by, but turbulence is extremely common everywhere in the world, such as when water flows across a rock in a stream or as one moves their hand through the air. It is near impossible to predict how a fluid undergoing turbulence will behave in the future (OpenStax, n.d.). For instance, it is impossible to predict where a specific molecule of water in a turbulent fluid will be some time in the future. However, if scientists were able to solve this problem, it would result in huge benefits across society: planes would drastically improve their efficiency as scientists would be able to know exactly how air behaves as it touches the wing. For the same reason, wind turbines would also become much more efficient and save billions of dollars (Neural Concept). Blood flow is an important example of turbulent flow and understanding it would allow doctors to prescribe medicine that flows to different parts of the body with a significant increase in accuracy (Discovering Engineering).

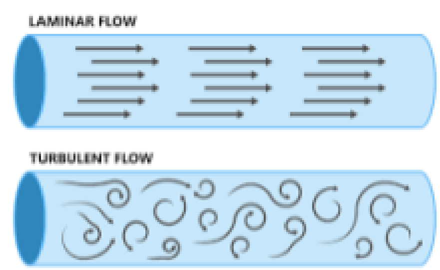


Fig. 1: Types of Fluid Flow

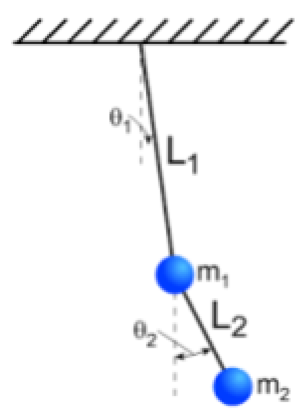


Fig 2. The Double Pendulum

This experiment centers around using chaos theory to understand this phenomenon. In a chaotic system, such as a double pendulum, releasing the pendulum from a specific point and then re-releasing it from a point even an atom's width away will result in the pendulum having two extremely different trajectories. To exactly predict the pendulum's motion, it is necessary to know the starting location with infinite precision, meaning infinite decimal places of accuracy. Similarly, in order to predict how the flow of a turbulent fluid will evolve in the future, it is necessary to know the velocity, temperature, and pressure at all points in the fluid with infinite accuracy, and even small differences in any one of any one of these variables initially can result in a vastly different fluid flow later on. This defines turbulence as a chaotic system, a system that is fully deterministic but is hypersensitive to initial conditions to the point that the system's evolution seems random. Turbulence's chaotic nature makes it difficult and even impossible to analyse using normal physics, but chaotic analysis may be more successful in finding patterns in its behavior. The experiment centered on obtaining position-time data from a particle undergoing turbulent motion in a boiling pot and analyzing this data using chaotic methods that will be explained in section two.

2. Related Work and Background

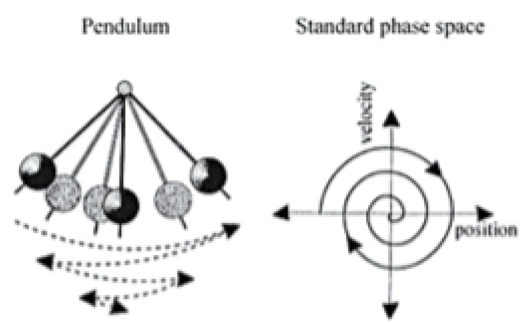


Fig 3. The Phase Space of a Pendulum

This experiment focused on finding the strange attractor in phase space of a particle undergoing turbulent motion, and the two key concepts from chaos theory used in this experiment were phase space and attractors. The phase space of the system is a plot containing all possible states of a system, and is one of the most important tools of chaotic analysis (Williams, 1997). A "state" of a system is the set of all attributes of the system at a specific point in time. Phase space is an essential part of chaos theory because it reveals hidden patterns in chaotic motion and displays relationships between the important variables of the system. For example, a pendulum has two physical attributes in its motion, velocity and position. The phase space of this pendulum is plotted in **Fig 2**, where velocity and position are plotted on the x and y axes respectively. As the pendulum is allowed to come to rest, states of the system are plotted in coordinate pairs and connected in time order with an opaque line, and each point in phase space fully describes the system at that point. This line is called the trajectory of the system. It is important to note that time does not directly appear on a phase space plot. It only appears indirectly through the trajectory.

2.1: Attractors

The other concept from chaos theory used in this paper is the strange attractor. An attractor is a set of states in phase space that the system tends towards (Williams, 1997). In **Fig 2**, the pendulum always ended up at the state where it had zero velocity and zero position(at the origin). This is an example of a point attractor, because the system was attracted to that specific point in phase space. However, this pendulum is a nonchaotic system. When a chaotic system makes an attractor, it is called a strange attractor, which is a complex three dimensional shape in phase space created when multiple trajectories that all started with slightly different initial conditions are all shown on the same phase plot. **Fig. 4** is the Lorentz attractor, a famous strange attractor that arises from a weather simulation plot. Even though the system never exactly repeats itself and is sensitive to tiny changes, it always traces paths within this butterfly shape when allowed to repeat from a different starting condition. This shows how chaos can have hidden order - the system is unpredictable but at the same time follows a broader pattern.

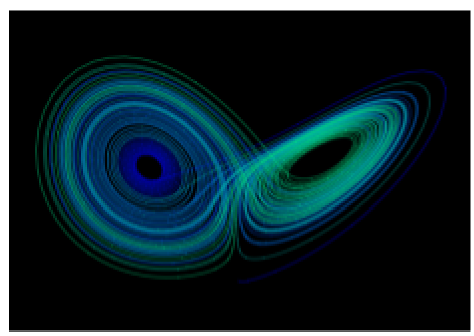


Fig. 4: The Lorenz Attractor

Attractors have potential to form whenever the system has “binding” characteristics, which are forces that tend to force the system into certain states(Williams, 1997). The system of the motion of a particle in boiling water has the capacity to form a strange attractor because of the tendency of hot and cold water to mix as the entropy of the system decreases. This mixing makes the system more likely to end up in some states than others, making the formation of a strange attractor in this system possible.

2.2: Experimental Setup and Research Questions

This experiment created a controlled turbulent environment by boiling water in a glass pot at a specific temperature on a gas stove. After this, a particle was released in the fluid while its motion was recorded in all three dimensions by two cameras recording in perpendicular dimensions. The particle was released in slightly different initial locations each time simply because it is impossible to release it in the same place every single time. Then, position-time data was extracted from the videos with motion tracking software and plotted in a 3D phase space with x position on one axis, y position on another, and total velocity on the z, connecting all points on the plot in time order with a line representing the trajectory of the system. Then, this was repeated thirty times to see if all thirty trajectories would form a strange attractor in phase space, again with the goal of uncovering hidden patterns in this seemingly random motion. The research questions that guided the research were:

- Will the motion of a particle in convection currents form a strange attractor in phase space?
- If a strange attractor is formed, can chaotic methods be done to further analyze the attractor and find patterns in the particle’s motion?

Research questions were considered to be more appropriate than hypotheses because of the nature of the experiment. The goal of the experiment is not to prove something right or wrong, but rather to see if a complex natural phenomenon can be understood when analyzed through a specific lens.

2.3: Related Work

No work was found that researches the exact same scenario as is covered in this paper. Other papers that are similar include the study "Chameleon Attractors in a Turbulent Flow," where Alberti et al. (2021) examined the attractors of turbulent fluids as a whole, not just the motion of a particle within them. Other papers, such as the "Motion of Inertial Particles in Gaussian Fields Driven by an Infinite-Dimensional Fractional Brownian Motion" by Schochel focused heavily on the mathematical theory behind turbulence.

3. Methodology & Procedure

1. Use food coloring to dye tapioca seeds red.
2. Mark the bottom of the glass pot using a marker.
3. Fill the pot with water up to an arbitrarily marked height on the side of the pot. It should be the most water the pot can hold without bubbling out while the water boils. Ensure that the same amount of water is present at the beginning of each trial.
4. Set up two cameras at right angles to each other, making sure that all three dimensions of motion in the water are captured.
5. Place the glass pot on the stove, making sure it is centered on the burner using the marks made earlier. Turn the heat to medium. Ensure that the pot is in the same orientation across all trials.

Fig. 5: The Experimental Setup

6. Place a thermometer probe into the water.
7. When the thermometer probe reaches 206 degrees, start recording on both videos at the same time
8. Take a fresh seed from the bag. Release the tapioca seed with tongs in the center of the pot.
9. Wait for two minutes, and then stop recording and remove the seed.
10. Repeat steps 1-9 thirty more times.

3.1: Data Analysis

Once all sixty videos are recorded (thirty from each camera), the analysis stage begins. The goal of this stage is to extract position-time data from the videos. For this step, Python code was used to track the motion of the particle in each video using OpenCV, clean the data, calculate velocity columns, and combine the Z position and velocity data from the side-view videos with the rest of the data to output thirty final CSV files containing a time column, position columns for all three dimensions, and velocity columns for all three dimensions. The Github link for this project can be found in the appendix. Around 85% of the code was generated by artificial intelligence, but extensive debugging was required.

3.2: Visualization

The CSV files were plotted in a three dimensional phase space with X position as the x axis, Y position as the y axis, and total velocity as the z-axis by the Python library Matplotlib. The total velocity was computed by summing up the individual velocity component columns with the Pythagorean theorem.

4. Results

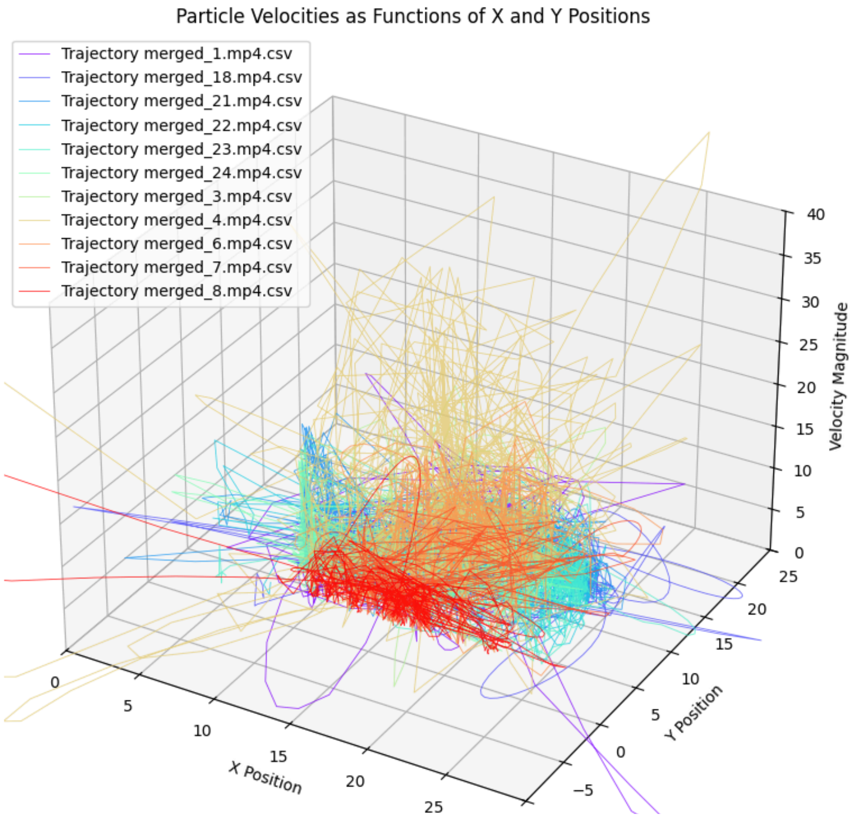
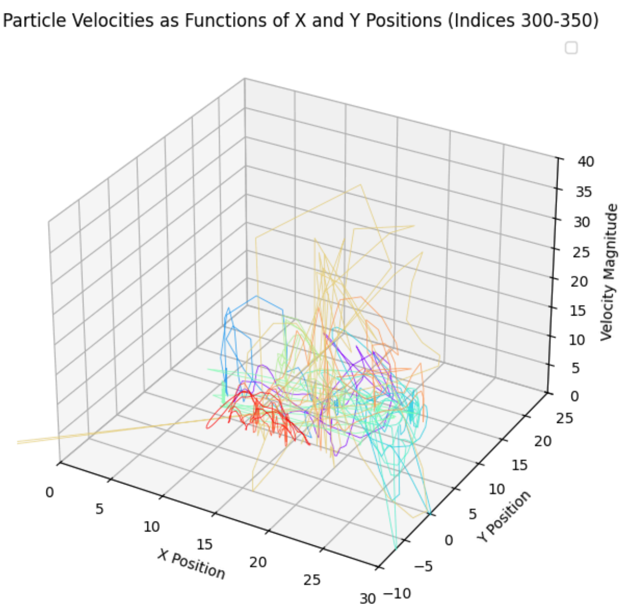


Fig. 6 - The trajectories of all ten particles in phase space with each colored line representing a different particle. Points(X position, Y position, Velocity Magnitude) were plotted for each particle and connected with a line in time order from the time the seed was dropped in the pot to two minutes afterwards.



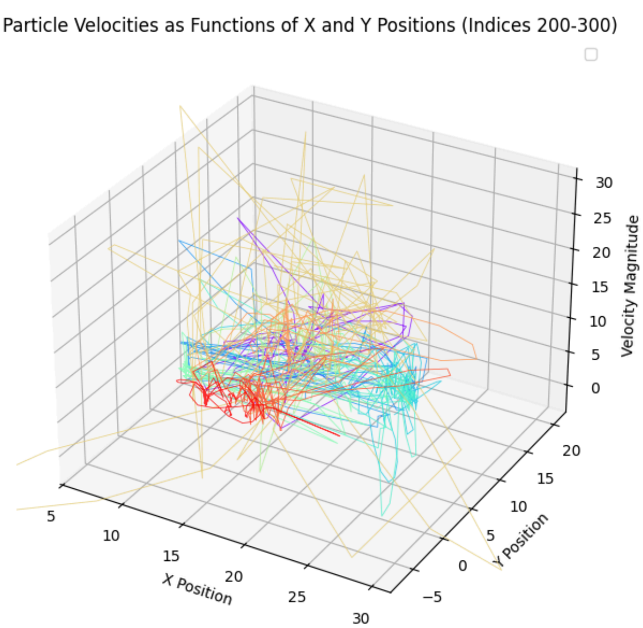


Fig 7 & Fig 8. These are both the exact same diagram as **Fig. 6**, except data for all ten particles is only shown over a seventeen-second timeframe and an eight-second time frame respectively, each starting and ending at an arbitrary time.

5: Discussion

Instead of the individual trajectories of each seed forming a defined three-dimensional shape in phase space, the phase space of the system showed random noise. There is no pattern or trend whatsoever in the data plots. The data is very crowded in **Figure 6**, but even when subsections of the data are examined by only displaying small durations in time, there is still no trend in the data. This result is unexpected because a strange attractor was predicted to form because of the binding nature of the system(see rationale). Thus, the data is inconclusive and no analysis can be done. Although it is possible that this experiment will not cause an attractor to appear in phase space, this random noise is likely caused due to experimental error.

5.1: Limitations

Firstly, the pot was much too small for the experiment, causing the seed to frequently run into the sides of the pot. This meant that the turbulence of the water had less of an effect on the motion of the particle because its motion was so often influenced by the force of glass on the seed. Using a larger pot would allow the motion of the particle to be less impeded by these boundary effects, and thus potentially show the strange attractor better. Second, the tapioca seeds sank when dropped in the pot. This meant that the seed interacted with the bottom of the pot before having a chance to really interact with the currents in the water, again obscuring the actual effect of the currents on the particle. Using a semi-buoyant particle instead would alleviate this problem. In addition, Open CV, the Python package that tracked the motion of the particle, could not track two-thirds of the videos because the camera went out of focus, making the video so blurry that it could not track it. This resulted in an enormous loss of data because of the low quality of the smartphone camera used.

6. Conclusion and Next Steps

The answer to the research question mentioned above is “No, the motion of a particle in convection currents will not form a strange attractor.” However, this conclusion should not be taken definitely because, as mentioned in the discussion, it may be entirely due to experimental error. If the data was conclusive, it would show that there is definitive order in the seeming randomness of turbulence, but it is not clear exactly how this would work because, again, the data is inconclusive. Possible next steps include repeating the experiment with the adjustments described above and possibly simulating the entire experiment using Computational Fluid Dynamics.

An attractor would have visually shown all the states the particle in the system tends towards, revealing patterns in how the particle's kinetic energy fluctuates over time. It would have been possible to quantify the exact degree of chaos in the system using Lyapunov exponents, approximating the time it takes for chaotic behavior to set in and revealing exactly how predictable or chaotic the system is. Additionally, the Fast Fourier Transform could have been used to find underlying frequencies in the motion of the particle and relate it to the geometry of the attractor. The formation of the strange attractor could have revealed important insights about how turbulent fluids fundamentally behave, insights that are at the cutting edge of current scientific understanding.

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Appendix

Github Link: https://github.com/AcheonDjon/pjas2025/tree/main